Autonomous RC Car: Starting Point for FSAE Driverless

What This Project Is

This is the starting point for building an autonomous version of our FSAE EV car. It isn't a semester project with a fixed scope. It's the foundation for something the team will be working on for years. The immediate goal is a small-scale autonomous RC car: a cheap, fast platform to prototype and fail on, long before any of this touches the real car.

The Autonomous Competition

Formula SAE has opened a new Driverless (DV) category, formally introduced for the 2026 season, with participation currently optional. It follows the model Formula Student Germany has run for years: the car has to complete a track with no driver and no pre-loaded map, using only what it can see and sense in real time.

The track is marked entirely by cones. Blue cones mark the left boundary, yellow cones mark the right. There's no course memorized ahead of time. The car has to detect the cones, figure out where the track actually goes, and plan a path through it as it drives.

SAE has published the official rules supplement and a roadmap for how the category will develop over the next several years. Both are worth reading in full before getting deep into design work:

https://www.fsaeonline.com/cdsweb/gen/DownloadDocument.aspx?DocumentID=cdd0c30e-662c-4c1a-b8a7-512b4e86f8ab

https://www.fsaeonline.com/cdsweb/gen/DownloadDocument.aspx?DocumentID=568eba8e-665f-4d1e-814b-d036f463c481

Why We're Starting With an RC Car

We are not touching the real car with any of this for a long time. A small-scale RC platform lets us prototype perception, localization, and path planning cheaply and iterate fast, without risking a car we can't afford to break. This is also how most other teams have approached driverless: build and prove out a scaled test platform first, then move to the real vehicle once the fundamentals actually work.

What This Actually Involves

There's no fixed spec for this yet, because we don't have all the answers yet either. Expect to spend real time researching each of these before committing to an approach:

- Sensors: what combination of cameras, LiDAR, IMU, and/or GPS actually works for cone detection and localization, and what we can realistically afford and integrate
- Simulation: how we test perception and planning without needing to run physical hardware every time. AMZ Driverless's open-source Formula Student Driverless Simulator (FSDS) is a good starting point to look into
- RC platform design: chassis, drivetrain, and how sensors and compute actually mount and power on a small scale

- Perception: detecting cones reliably, likely with computer vision or LiDAR-based object detection. The FSOCO dataset (fsoco-dataset.com) is an open, annotated cone dataset a lot of teams train on instead of building their own from scratch
- Localization and mapping: figuring out where the car is and where the track boundaries are in real time, without a pre-loaded map
- Path planning: turning detected cones into a safe, fast path through the track

Reading what other teams have already published is part of the job, not a detour from it. AMZ Driverless and MIT Driverless both have well-known whitepapers on their approaches, and they're a much better starting point than reinventing this from nothing.

Timeline and Commitment

This is a multi-year effort. Expect more than a year of research, prototyping, and iteration on the RC platform before any of this goes anywhere near the actual EV car.

If you want to help build the foundation other members will be racing on in a few years, this is it.

This Is a Collaborative Effort

This isn't something one person figures out alone, and it isn't going to be solved by one semester's new member class either. It will involve people across electrical, mechanical, and software, building on each other's work across multiple semesters. Document what you learn as you go, the same way the whitepapers you'll be reading were written, so the next person doesn't start from zero.

Getting Started

Talk to EV Electrical about where this stands right now and how to get involved. Before doing any design work, read through the official rules supplement and roadmap, look at what AMZ and MIT Driverless have published, and get familiar with the FSOCO dataset. Come with a sense of where you actually want to contribute, whether that's sensors, software, or the RC platform itself.